Student Takeaway for Sync Session

<<Week 5: Programming Fundamentals for Machine Learning – SQL>>

This guide is your roadmap to making the most of our online session. Packed with essential tips and strategies, it's designed to keep you engaged, prepared, and ready to dive into a smooth and productive learning journey. Get ready to participate, learn, and thrive!

Session Overview

<table>
<tr><td>Session title</td><td>Review Of Basics Of Mathematical Topics, and
Programming Fundamentals for Machine Learning – SQL</td></tr>
<tr><td>Session duration</td><td>2 hours</td></tr>
<tr><td>Session type</td><td>Live Virtual Session</td></tr>
<tr><td>Scope</td><td>This session introduces key statistical and probability concepts used in data science. Topics covered include:
Hypothesis testing and statistical inference.
Real-world applications in decision-making and analytics.
Advanced SQL for effective data handling.
Build and query relational tables</td></tr>
<tr><td>Learning objectives</td><td><table>
<tr><th>Objective</th><th>Core capability</th></tr>
<tr><td>Perform hypothesis testing using statistical tools</td><td>Practical application in decision-making</td></tr>
<tr><td>Set up and interact with SQL databases and tables</td><td>Data Management & Query Writing</td></tr>
<tr><td>Perform complex queries using joins, variables, functions, and stored procedures</td><td>Data Analysis and Retrieval</td></tr>
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<tr><td>Software/Tools</td><td>MySQL or equivalent SQL database (phpMyAdmin, MySQL Workbench)</td></tr>
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Pro tips for success:

- **Type Along:** Practice queries as we go for muscle memory.
- **Tweak and Try:** Change inputs, try new columns, and see what changes.
- **Debug Together:** Ask questions—error messages are clues!
- **Link Concepts:** Think about how SQL links to your ML projects.

Session Details

Topic/Issue	What to expect	Notes
Hypothesis Testing	Learn null vs. alternative hypotheses and statistical significance	How do we validate claims using hypothesis tests?
t-Test & Z-Test	Differentiate between small and large sample tests	When should you use a t-test vs. a z-test?

Topic/Issue	What to expect	Notes
ANOVA & Chi-Square Test	Compare multiple groups and categorical data relationships	How do businesses use statistical tests?
Random Variables	Understanding discrete and continuous random variables	How do random variables impact probability modeling?
Discrete Probability Distributions	Learn probability mass functions and their uses	Applications in product testing and surveys
Poisson Distribution	Predict the probability of rare events	How does Poisson help in traffic analysis?
Continuous Probability Distributions	Explore probability density functions	Why is continuous probability
Central Limit Theorem	Learn how sampling distributions work	Why is CLT fundamental in statistics?
Confidence Intervals	Understand margin of error and statistical estimation	How do confidence intervals help in research?
Hypothesis Testing Deep Dive	Learn about p-values and significance levels	What does a p-value of 0.05 mean?
Errors in Hypothesis Testing	Distinguish between Type I and Type II errors	Which error is worse in medical testing?
Choosing the Right Statistical Test	Guidelines for selecting the appropriate test	When do you use an ANOVA vs. a chi-square test?
Introduction	Session goals and agenda overview	Why SQL matters in ML
Setting Up Databases and Tables	Creating and using databases and tables	Syntax for MySQL
Inserting Data	Adding records into tables	Real-world employee data
SQL Joins	Combining data across tables	Inner join between employees and departments
Variables and Functions	Use of variables, calculations	Salary increase example
Subqueries and Views	Building reusable queries	Average salary subquery and high-paid employees view
String and Date-Time Functions	Data formatting and calculations	UPPER function, experience calculation
Stored Procedures	Efficient querying with parameters	Procedure to fetch employees by department
Advanced SQL Techniques	CTEs, Indexing, Grouping, ROLLUP	Performance optimisation examples

Post session

Reflection challenge	What did you find exciting about decision trees?
Explore more	• Try writing SQL queries on practice platforms like SQLZoo or LeetCode (Database section). • Watch a short tutorial on SQL Joins or Views to reinforce your understanding**.**
Get inspired	SQL is your key to understanding data. With joins, views, and subqueries, you're now equipped to uncover insights that power better models. Stay curious—great ML starts with great data.
The journey ahead	In the next session, you'll dive into data preprocessing and feature engineering—learning to clean, transform, and enhance datasets using pandas and scikit-learn to prepare for effective model building.